

2023-Paper

m -dependent Potential

$$V_m^-(x) = V(x) - 2 \left[\frac{\mathcal{H}_m''}{\mathcal{H}_m} - \left(\frac{\mathcal{H}_m'}{\mathcal{H}_m} \right)^2 + 1 \right], \quad (1)$$

$$\psi_{0,m}^-(x) = \left(\frac{2^m m!}{\sqrt{\pi}} \right)^{\frac{1}{2}} \frac{e^{-\frac{x^2}{2}}}{\mathcal{H}_m(x)}, \quad (2)$$

$$\psi_{n+1,m}^-(x) = \frac{1}{\sqrt{E_{n,m}^+ 2^n n! \sqrt{\pi}}} \frac{e^{-\frac{x^2}{2}}}{\mathcal{H}_m(x)} y_{n+1}^m(x), \quad n = 0, 1, 2, \dots, \quad (3)$$

$$E_{n+1,m}^- = 2(n + m + 1), \quad n = 0, 1, 2, \dots, \quad \text{and } E_{0,m}^- = 0. \quad (4)$$

m and λ -dependent Potential

$$\hat{V}_m^-(\lambda, x) = V_m^-(x) - 2 \frac{d^2}{dx^2} \ln [\mathcal{I}_m(x) + \lambda], \quad (5)$$

$$\mathcal{I}_m(x) = \left(\frac{2^m m!}{\sqrt{\pi}} \right) \int_{-\infty}^x \left[\frac{e^{-\frac{x'^2}{2}}}{\mathcal{H}_m(x')} \right]^2 dx'. \quad (6)$$

$$\hat{\psi}_{0,m}^-(\lambda, x) = \sqrt{\lambda(1 + \lambda)} \frac{\psi_{0,m}^-(x)}{\mathcal{I}_m(x) + \lambda}. \quad (7)$$

$$\begin{aligned} \hat{\psi}_{n+1,m}^-(\lambda, x) &= \frac{1}{E_{n+1,m}^-} \hat{A}^\dagger A \psi_{n+1,m}^-(x), \quad n = 0, 1, 2, \dots, \\ &= \frac{e^{-\frac{x^2}{2}}}{\sqrt{2(n + m + 1) 2^n n! \sqrt{\pi}}} \left(\frac{y_{n+1}^m(x)}{\mathcal{H}_m(x)} + H_n(x) \frac{d}{dx} \ln [\mathcal{I}_m(x) + \lambda] \right) \end{aligned} \quad (8)$$

When $\lambda = 0$ or 1 (P or AM) the spectrum is strictly isospectral to the spectrum of starting potential.

$$E_{n,m}^+ = 2(n + m + 1), \quad n = 0, 1, 2, \dots \quad (9)$$

2025-Paper

$$\left[-\frac{d^2}{dx^2} + V(x) \right] \Psi = E\Psi \quad \text{where} \quad V(x) = \frac{1}{4}\omega_1^2 x^2. \quad (10)$$

$$V^l(x) = V(x) + \lambda_0 x, \quad (11)$$

$$\tilde{V}^l(\tilde{x}) = \frac{1}{4}\omega_1^2 \tilde{x}^2 - \frac{\lambda_0^2}{\omega_1^2} \quad \text{where} \quad \tilde{x} = x + \frac{2\lambda_0}{\omega_1^2}, \quad (12)$$

$$E \rightarrow \tilde{E}_n^l(\omega_1) = \left(n + \frac{1}{2} \right) \omega_1 + \frac{\gamma_0^2}{\omega_1^2}; \quad n = 0, 1, 2, \dots \quad (13)$$

$$\tilde{\psi}_n^l(\tilde{x}) = \psi_n(x), \quad \text{Here,} \quad \tilde{\omega}_1 = \omega_1. \quad (14)$$

$$\mathcal{PT} \tilde{\psi}_n^l(\tilde{x}) = (-1)^n \tilde{\psi}_n^l(\tilde{x}) \quad (15)$$

1D

$$V_{RE,m}(x) = V(x) + V_{rat,m}(x), \quad -\infty < x < \infty \quad (16)$$

$$V_{rat,m}(x) = -2 \left[\frac{\mathcal{H}_m''(\sqrt{\frac{\omega_1}{2}}x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} - \left[\frac{\mathcal{H}_m'(\sqrt{\frac{\omega_1}{2}}x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \right]^2 + \frac{\omega_1}{2} \right] \quad (17)$$

$$\begin{aligned} \psi_{RE,m,0}(x) &= \frac{\zeta(\omega_1, x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \\ \text{and } \psi_{RE,m,n+1}(x) &= \frac{\zeta(\omega_1, x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \hat{H}_{m,n+1} \left(\sqrt{\frac{\omega_1}{2}}x \right), \end{aligned} \quad (18)$$

$$\hat{H}_{m,n+1}(x) = \mathcal{H}_m(x)H_{n+1}(x) + H_n(x)\frac{d}{dx}\mathcal{H}_m(x), \quad (19)$$

$$E_{RE,m,n+1}(\omega_1) = (n + m + 1)\omega_1 \quad \text{with} \quad E_{RE,m,0}(\omega_1) = 0. \quad (20)$$

$$\tilde{V}_{RE,m}^l(\tilde{x}) = \tilde{V}^l(\tilde{x}) + V_{rat,m}(\tilde{x}), \quad -\infty < x < \infty \quad (21)$$

$$\begin{aligned} \tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \frac{\zeta(\omega_1, \tilde{x})}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}\tilde{x})} \\ \text{and } \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= \frac{\zeta(\omega_1, \tilde{x})}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}\tilde{x})} \hat{H}_{m,n+1} \left(\sqrt{\frac{\omega_1}{2}}\tilde{x} \right). \end{aligned} \quad (22)$$

$$\tilde{E}_{RE,m,n+1}^l(\omega_1) = (n+m+1)\omega_1 \quad \text{with} \quad \tilde{E}_{RE,m,0}^l(\omega_1) = 0. \quad (23)$$

$$\begin{aligned} \mathcal{PT} \tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \tilde{\psi}_{RE,m,0}^l(\tilde{x}) \\ \text{and } \mathcal{PT} \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= (-1)^{n+1} \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) \end{aligned} \quad (24)$$

2D

$$V(x, y) = \frac{1}{4}\omega_1^2 x^2 + \frac{1}{4}\omega_2^2 y^2 + \frac{\lambda}{2}xy. \quad (25)$$

$$\begin{aligned} x &= a\tilde{x} + b\tilde{y}, \\ y &= -b\tilde{x} + a\tilde{y}, \end{aligned} \quad \begin{cases} \tilde{x} = ax - by, \\ \tilde{y} = bx + ay, \end{cases} \quad (26)$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\frac{1}{2} \left(\omega_1^2 + \omega_2^2 - \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \right)} \\ \text{and } \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega_1^2 + \omega_2^2 + \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \right)} \end{aligned} \quad (27)$$

$$\tilde{V}(\tilde{x}, \tilde{y}) = \tilde{V}(\tilde{x}) + \tilde{V}(\tilde{y}). \quad (28)$$

$$\tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) \propto \tilde{\psi}_{n_1}(\tilde{x}) \tilde{\psi}_{n_2}(\tilde{y}), \quad n_1, n_2 = 0, 1, 2, \dots, \quad (29)$$

$$\text{and } \tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2) = \left(n_1 + \frac{1}{2}\right)\tilde{\omega}_1 + \left(n_2 + \frac{1}{2}\right)\tilde{\omega}_2, \quad (30)$$

$$\sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \leq \omega_1^2 + \omega_2^2, \quad (31)$$

$$\begin{aligned} \sqrt{-4\gamma^2 + (\omega_1^2 - \omega_2^2)^2} &\leq \omega_1^2 + \omega_2^2, \\ \text{and } -4\gamma^2 + (\omega_1^2 - \omega_2^2)^2 &> 0. \end{aligned} \quad (32)$$

$$\eta = \begin{pmatrix} -k & -\sqrt{1-k^2} \\ \sqrt{1-k^2} & -k \end{pmatrix} \quad (33)$$

$$V^\dagger = \eta V \eta^{-1}. \quad (34)$$

0.1

$$\tilde{V}_{RE,m_1,m_2}(\tilde{x}, \tilde{y}) = \tilde{V}_{RE,m_1}(\tilde{x}) + \tilde{V}_{RE,m_2}(\tilde{y}), \quad (35)$$

$$\tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x}, \tilde{y}) \propto \tilde{\psi}_{RE,m_1,0}(\tilde{x}) \tilde{\psi}_{RE,m_2,0}(\tilde{y}) \quad (36)$$

$$\tilde{\psi}_{RE,m_1,m_2,n_1+1,n_2+1}(x, y) \propto \tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x}) \tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y}) \quad (37)$$

$$\text{and } \tilde{E}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{\omega}_1, \tilde{\omega}_2) = \tilde{E}_{RE,m_1,n_1+1}(\tilde{\omega}_1) + \tilde{E}_{RE,m_2,n_2+1}(\tilde{\omega}_2) \quad (38)$$

$$\text{with } \tilde{E}_{RE,m_1,m_2,0,0}(\tilde{\omega}_1, \tilde{\omega}_2) = 0, \quad n_1, n_2 = 0, 1, 2, \dots, \dots,$$

0.2

$$\frac{\tilde{\omega}_1}{\tilde{\omega}_2} = \frac{\tilde{r}}{\tilde{s}}, \quad (39)$$

$$\tilde{V}(\tilde{x}, \tilde{y}) = \frac{1}{4} (\tilde{\omega}_1^2 \tilde{x}^2 + \tilde{\omega}_2^2 \tilde{y}^2). \quad (40)$$

$$\lambda(r, s, \tilde{r}, \tilde{s}, \omega_1, \omega_2) = \frac{\sqrt{(\tilde{r}^4 + \tilde{s}^4) - \left(\frac{\tilde{r}\tilde{s}}{rs}\right)^2 (r^4 + s^4)}}{\tilde{r}^2 + \tilde{s}^2} \omega_1 \omega_2 \quad (41)$$

$$\tilde{E}_{n_1,n_2}(\tilde{r}, \tilde{s}, \tilde{\omega}_2) = \left(\tilde{r}n_1 + \tilde{s}n_2 + \frac{\tilde{r} + \tilde{s}}{2} \right) \frac{\tilde{\omega}_2}{\tilde{s}}. \quad (42)$$

0.2.1

$$\begin{aligned} \tilde{x} &= \frac{1}{\sqrt{2}} (x + y), \\ \tilde{y} &= \frac{1}{\sqrt{2}} (y - x), \end{aligned} \quad \begin{cases} \tilde{\omega}_1 = \sqrt{\omega^2 - \lambda}, \\ \tilde{\omega}_2 = \sqrt{\omega^2 + \lambda}. \end{cases} \quad (43)$$

$$\lambda(\tilde{r}, \tilde{s}, \omega) = \frac{|\tilde{r}^2 - \tilde{s}^2|}{\tilde{r}^2 + \tilde{s}^2} \omega^2. \quad (44)$$

$$E_{\lambda=0} = (n_1 + n_2 + 1) \omega \quad (45)$$

$$V(x, y) = \frac{1}{4} \left(\frac{1}{2} x^2 + \frac{9}{2} y^2 \right) + \frac{\sqrt{7}}{4} xy. \quad (46)$$

0.2.2

$$V(x, y) = \frac{1}{4} (x^2 + 9y^2) + i \frac{\sqrt{7}}{2} xy. \quad (47)$$

3D

$$V^{lq}(x, y, z) = \frac{1}{4} (\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2) + \lambda_0 z + \frac{\lambda}{2} xy. \quad (48)$$

$$V^{lq}(x, y, z) = V(x, y) + V^l(z), \quad (49)$$

$$\tilde{V}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}(\tilde{x}, \tilde{y}) + \tilde{V}^l(\tilde{z}). \quad (50)$$

$$\tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) \tilde{\psi}_{n_3}^l(\tilde{z}) \quad (51)$$

$$\tilde{E}_{n_1, n_2, n_3}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{n_3}^l(\tilde{\omega}_3) \quad (52)$$

$$\mathcal{P}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \quad (53)$$

$$\mathcal{P}_2 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = (-1)^{n_3} \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}).$$

$$\mathcal{P}_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad \mathcal{P}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (54)$$

$$\mathcal{P}_1 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \mathcal{P}_3 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = (-1)^{n_1+n_2} \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}).$$

$$\mathcal{P}_4 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad (55)$$

$$\mathcal{P}_4 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = (-1)^{n_1+n_2+n_3} \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}).$$

0.2.3

$$\tilde{V}_{RE, m_1, m_2, m_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y}) + \tilde{V}_{RE, m_3}^l(\tilde{z}) \quad (56)$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, 0, 0, 0}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, m_2, 0, 0}(\tilde{x}, \tilde{y}) \tilde{\psi}_{RE, m_3, 0}^l(\tilde{z}), \quad (57)$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{x}, \tilde{y}) \tilde{\psi}_{RE, m_3, n_3+1}^l(\tilde{z}) \quad (58)$$

$$\tilde{E}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{RE, m_3, n_3+1}^l(\tilde{\omega}_3) \quad (59)$$

$$\tilde{E}_{RE, m_1, m_2, m_3, 0, 0, 0}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{RE, m_1, m_2, 0, 0}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{RE, m_3, 0}^l(\tilde{\omega}_3)$$

0.3

$$V(x, y, z) = \frac{1}{4} (\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda_2 yz + 2\lambda_3 zx). \quad (60)$$

$$V^q(x, z, y) = \frac{1}{4} (\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda_2 yz + 2\lambda_3 zx). \quad (61)$$

0.3.1

$$V^{q_1}(x, y, z) = \frac{1}{4} (\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_2 yz + 2\lambda_3 zx) \quad (62)$$

$$\begin{cases} x = ad\tilde{y} + bd\tilde{z} - c\tilde{x}, \\ y = d\tilde{x} + ac\tilde{y} + bc\tilde{z}, \\ z = a\tilde{z} - b\tilde{y}, \end{cases} \begin{cases} \tilde{x} = -cx + d y, \\ \tilde{y} = ad x + acy - bz, \\ \tilde{z} = bd x + bcy + az. \end{cases} \quad (63)$$

$$\tilde{V}^q(\tilde{x}, \tilde{y}, \tilde{z}) = \frac{1}{4} (\tilde{\omega}_1^2 \tilde{x}^2 + \tilde{\omega}_2^2 \tilde{y}^2 + \tilde{\omega}_3^2 \tilde{z}^2) \quad (64)$$

$$\begin{aligned} \tilde{\omega}_1 &= \omega, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 - \sqrt{4(\lambda_2^2 + \lambda_3^2) + (\omega^2 - \omega_3^2)^2} \right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \sqrt{4(\lambda_2^2 + \lambda_3^2) + (\omega^2 - \omega_3^2)^2} \right)}. \end{aligned} \quad (65)$$

$$\tilde{V}_{RE, m_1, m_2, m_3}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}_{RE, m_1}(\tilde{x}) + \tilde{V}_{RE, m_2}(\tilde{y}) + \tilde{V}_{RE, m_3}(\tilde{z}) \quad (66)$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, 0, 0, 0}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, 0}(\tilde{x}) \tilde{\psi}_{RE, m_2, 0}(\tilde{y}) \tilde{\psi}_{RE, m_3, 0}(\tilde{z})$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, n_1+1}(\tilde{x}) \tilde{\psi}_{RE, m_2, n_2+1}(\tilde{y}) \tilde{\psi}_{RE, m_3, n_3+1}(\tilde{z}) \quad (67)$$

$$\begin{aligned} \tilde{E}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{q_1}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) &= \tilde{E}_{RE, m_1, n_1+1}(\tilde{\omega}_1) \\ &\quad + \tilde{E}_{RE, m_2, n_2+1}(\tilde{\omega}_2) + \tilde{E}_{RE, m_3, n_3+1}(\tilde{\omega}_3) \end{aligned} \quad (68)$$

$$\tilde{E}_{RE, m_1, m_2, m_3, 0, 0, 0}^{q_1}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = 0 \quad (69)$$

$$0 \leq \lambda_2^2 + \lambda_3^2 \leq \omega^2 \omega_3^2 \quad (70)$$

$$\sqrt{\lambda_2^2 + \lambda_3^2} = \frac{\sqrt{(\tilde{u}^4 + \tilde{v}^4) - \left(\frac{\tilde{u}\tilde{v}}{uv}\right)^2 (u^4 + v^4)}}{\tilde{u}^2 + \tilde{v}^2} \omega \omega_3 \quad (71)$$

$$\sqrt{\lambda_2^2 + \lambda_3^2} = \frac{\sqrt{2(\tilde{u}^2 - \tilde{v}^2)^2 - (\tilde{u}\tilde{v})^2}}{\tilde{u}^2 + \tilde{v}^2} \omega^2 \quad (72)$$

0.3.2

$$V^{q_2}(x, y, z) = \frac{1}{4} \left(\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda(yz + zx) \right). \quad (73)$$

$$\begin{aligned} x &= \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} - \frac{\tilde{x}}{\sqrt{2}}, \\ y &= \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} + \frac{\tilde{x}}{\sqrt{2}}, \\ z &= a\tilde{z} - b\tilde{y}, \end{aligned} \quad \begin{cases} \tilde{x} = \frac{y}{\sqrt{2}} - \frac{x}{\sqrt{2}}, \\ \tilde{y} = \frac{a}{\sqrt{2}}x + \frac{a}{\sqrt{2}}y - bz, \\ \tilde{z} = \frac{b}{\sqrt{2}}x + \frac{b}{\sqrt{2}}y + az. \end{cases} \quad (74)$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\omega^2 - \lambda_1}, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \lambda_1 - \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2} \right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \lambda_1 + \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2} \right)}. \end{aligned} \quad (75)$$

$$\begin{aligned} -\omega^2 &\leq \lambda_1 \leq \omega^2 \\ \text{and } 0 &\leq \lambda_1^2 \leq \frac{(\omega^2 + \lambda)\omega_z^2}{2}. \end{aligned} \quad (76)$$

$$-\omega^2 \leq \lambda_1 \leq \omega^2 \quad \text{and} \quad 0 \leq \gamma^2 \leq \frac{(\omega^2 - \omega_3^2 + \lambda_1)^2}{8} \quad (77)$$

$$\sqrt{8\lambda^2 + \lambda_1^2} = \frac{|\tilde{v}^2 - \tilde{u}^2|}{\tilde{v}^2 + \tilde{u}^2} (2\omega^2 + \lambda_1), \quad (78)$$

$$\lambda = \frac{\sqrt{\left(\frac{|\tilde{v}^2 - \tilde{u}^2|}{\tilde{v}^2 + \tilde{u}^2} (\lambda_1 + 2\omega^2) \right)^2 - \lambda_1^2}}{2\sqrt{2}}. \quad (79)$$